Fiscal Year:	FY 2012	**Task Last Updated:**	FY 03/19/2012
PI Name:	Ott, C. Mark Ph.D.		
Project Title:	Microbial Risk Assessment		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Environmental health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	(1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response (2) **Microhost**:Risk of Adverse Health Effects Due to Host-Microorganism Interactions		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2008	**End Date:**	10/01/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/01/2011 per PI (Ed., 11/17/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mena, Kristina (University of Texas Health Science Center at Houston)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To mitigate health risks to the crew, the JSC Microbiology Laboratory establishes health standards and requirements. Historically, microbiological spaceflight requirements have been established in a subjective manner based upon expert opinion of both available environmental and clinical monitoring results and the incidence of disease. Current microbiology requirements were based on Apollo, Skylab, and early shuttle data. The use of out-of-date requirements could unnecessarily increase costs/resource usage and provide inadequate or inappropriate data for health assessment. The goal of the Microbial Risk Assessment Study is to utilize a systematic approach toward evaluation of current data, identify gaps in knowledge, and develop confidence levels, identify risks, and determine acceptable and unacceptable risks. The study will use quantitative microbial risk assessment approaches that have been applied to great success in the food and potable water industries. The application of these methods to latest data sets obtained in space travel will allow the most appropriate data sets to be assessed for health assessment. Finding are expected to lead to the development of a systematic approach that will reduce health risks to crews of ISS, lunar sortie, lunar habitation, and Mars missions while at the same time reducing costs by optimizing allocation of resources.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Microbial Risk Assessments are performed by numerous government agencies, such as the EPA and USDA, municipal water production facilities, and the food industry. The use of these techniques to the semiclosed environment of the International Space Station provides unique insight into changes in microbiota and transfer of specific organisms between individuals that can be translated to risk assessments for the general population on Earth.
Task Progress:	The objective of this study was to develop and apply the quantitative microbial risk assessment (QMRA) framework at the International Space Station (ISS). Specifically, QMRA was used to translate ISS microbial water monitoring data to flight crew health impact in order to direct water monitoring practices and inform health risk reduction strategies. QMRA is a four-step paradigm – including (1) hazard identification; (2) dose-response assessment; (3) exposure assessment; and, (4) risk characterization – that is used to evaluate human health risks resulting from environmental exposure to pathogens. For this project, the framework was reviewed and modified for its applicability to address flight crew health risks associated with exposure to potable water at the ISS. The research plan was divided into four (4) phases. Phase 1 involved the initial evaluation of the QMRA process with the objective of identifying critical areas where QMRA can be effectively applied to the spaceflight scenario as well as areas where data are needed. Potable water monitoring data for microorganisms were reviewed and potential waterborne pathogens at the ISS were identified and characterized. Phase 2 specifically addressed the dose-response step of the QMRA process. Using microbial water monitoring data obtained at the ISS, dose-response relationships that represent the flight crew (host) and microorganism (potential pathogen) interaction were developed. This Phase of the project identified Pseudomonas as an appropriate model microorganism to address potable water quality at the ISS. Pseudomonas was chosen as a microbe of interest because of its occurrence in ISS water, the availability of dose-response data derived from human volunteer studies and published in the peer-reviewed literature, and its association with a wide range of human health effects. Pseudomonas offers unique challenges because it is ubiquitous in the environment, is an opportunistic pathogen, is resistant to different types of disinfectants, is capable of forming biofilms, and has the ability to grow in unusual environments. In addition, in some ways Pseudomonas is representative of several members of the heterotrophic plate count (HPC) bacteria, which are routinely detected in ISS potable water. Furthermore, Pseudomonas is transmitted through water through a variety of different exposure routes, which is a necessary consideration when addressing flight crew exposures to water at the ISS. Phase 3 focused on the exposure assessment aspect of the QMRA. Several pertinent questions related to exposure were initially identified and included: (1) What factors affect the occurrence of pathogenic microbes at ISS?; (2) What factors affect the likelihood of flight crew exposure to waterborne microbes at ISS?; (3) What factors affect the extent of flight crew exposure to waterborne microbes at ISS?; (4) In what ways are flight crew exposed to waterborne microbes at ISS?; (5) How important are certain exposure factors to flight crew health risks?; and, (6) How do estimated risks associated with ISS potable water compare to health risks in terrestrial water environments? Previous QMRAs involving the terrestrial water environment have shown that it is the exposure assessment step that contributes the greatest source of variability and uncertainty in the QMRA. Typically, exposure-related input parameters are either unknown (thereby assumptions are employed) or they represent a wide-range of plausible values. This Phase identified key aspects of the ISS environment that introduce different challenges related to exposure, which can subsequently impact health risks. This more recent Phase of the QMRA project (Phase 4) integrated all of the information learned from the first three Phases to complete a risk characterization of flight crew exposure to identified hazards in ISS potable water. The approach was both qualitative and quantitative, with the overall objective to predict flight crew health outcomes from exposure to water at the ISS. Infection risk ranges estimated in Phase 3 for Pseudomonas spp., Salmonella and E. coli were evaluated for practical implications. Typically, the water industry concentrates research and monitoring efforts on improving quantification of microorganisms in finished drinking water. This is significant because it can be challenging to develop and apply field and laboratory methodologies that are sensitive enough to adequately enumerate the small amount of pathogens present in drinking water that may negatively impact the health of consumers. Even minimal changes in microbial concentration as seen in finished drinking water supplies have been shown to dramatically alter risk. Other studies have demonstrated that pathogens undergo virulence changes during spaceflight, and some dysfunction may occur in the crew immune system. As such, this makes dose-response characterization a top priority. Furthermore, identifying which microorganisms may potentially pose a health threat to a spaceflight crew can also direct mitigation strategies that may involve disinfection regimes with specific microbial targets and/or flight crew treatment using appropriate antibiotics. A manuscript is being prepared describing the Microbial Risk Assessment Study.
Bibliography Type:	Description: (Last Updated: 10/03/2025)